

Forecasting solar power in Canton Lucerne

Classical SARIMAX vs. a time-series foundation model

Switzerland is expanding solar power fast, but solar output swings with weather and season, which makes the grid harder to plan. We forecast hourly solar production with an explainable SARIMAX model on open MeteoSwiss weather data and test whether the time-series foundation model TimesFM can beat it.

17,205

PV plants in operation, Lucerne [1]

647 MW

installed PV capacity, Lucerne [1]

SDG 7.2

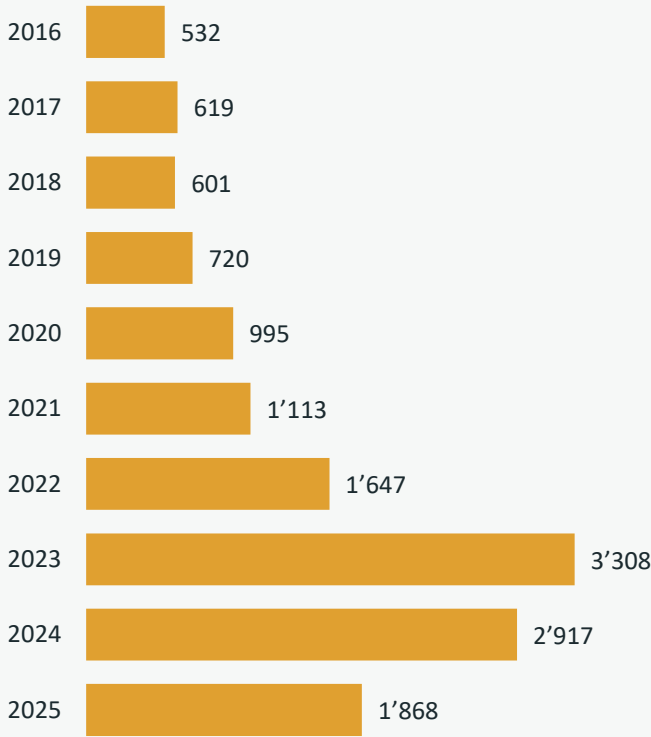
raise the share of renewable energy [11]

Open data we can build on

Source	What it contains	Role in the project
[5] Energy-Charts (Fraunhofer ISE)	Hourly Swiss solar generation; usable from 2018	Forecast target
[4] MeteoSwiss Open Data, station Luzern	Radiation, temperature, sunshine; 10-min and hourly	Weather covariates (the X in SARIMAX)
[1] SFOE register of power plants	17,205 PV plants in Lucerne: kW, start date, tilt, orientation	Installed capacity over time
[2] SFOE solar potential per municipality	Suitable roof and facade potential	Untapped potential (extension)
[3] SFOE study “Winter power from PV”	PV production profiles of all municipalities, 2004–2018	Validation, winter share (extension)

Open question: forecast the measured national series [5], or a Lucerne series modelled from installed capacity [1] and local radiation [4] with pvlib [9]?

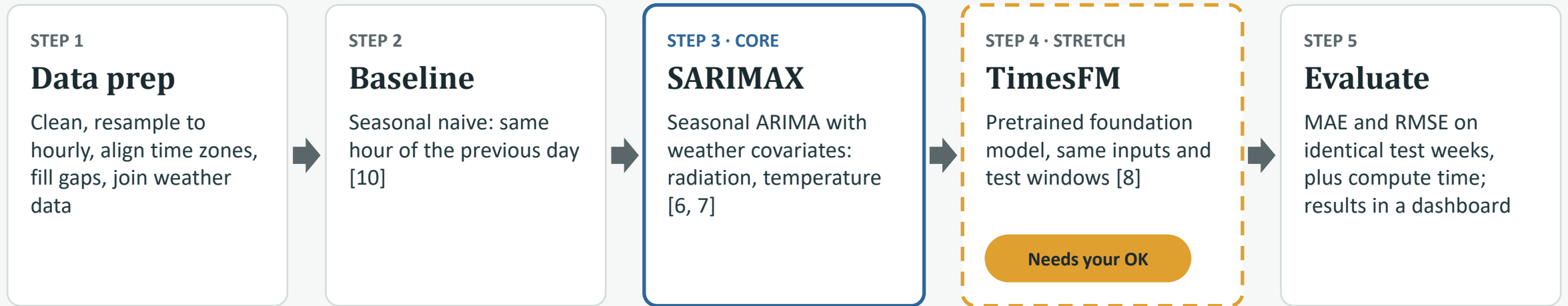
New PV plants per year, Canton Lucerne



Plants in operation, by start year. Own count from [1].

Sources: [1]–[5], [9], see References. All checked on 8 Oct 2026.

Approach: same data prep, two models



Sustainability angle (SDG 7.2) [11]

Better solar forecasts make PV easier to integrate into the grid [12]: heat pumps, EV charging and batteries can be shifted into sunny hours.

Reflection and extensions

Is a large model worth its extra compute for the accuracy it adds? If time allows: more cantons, untapped potential [2], winter share [3].

Sources: [6] Shumway & Stoffer 2017 · [7] Seabold & Perktold 2010 · [8] Das et al. 2024 · [10] Hyndman & Athanasopoulos 2021 · [11] UN · [12] Swissgrid

First tasks and questions for you

Until	Task
15 Oct	D1: project title and pitch in the README (pull request)
22 Oct	D2: problem statement and SMART goals
End of Oct	Load and check the data, set up the environment, first EDA notebook
12 Nov	D3: data sources and EDA

Proposed split

Elias: data sourcing and modelling · Flurin: analysis and visualisation

Questions for you

1. Is TimesFM fine as a benchmark next to SARIMAX, or out of scope given the course's "no foundation models" rule?
2. One focused research question, or may we add related ones (untapped potential, winter share)?
3. Forecast target: the measured national solar series or a modelled Lucerne series?
4. Start with Canton Lucerne and extend to other cantons later?
5. Our group number for the repo name (dspro01_hs26_groupXX)?

Deadlines from the course template README (github.com/curdon/dspro01_hs26_template).

References

Data

- [1] Swiss Federal Office of Energy (SFOE). (2026). *Elektrizitätsproduktionsanlagen* [Data set]. opendata.swiss. Updated 24.09.2026.
- [2] SFOE. (2026). *Solarenergiepotenziale der Schweizer Gemeinden – Dächer und Fassaden* [Data set]. bfe-ogd.ch.
- [3] SFOE. (2019). *Studie Winterstrom aus Photovoltaik – Produktionsprofile aller Schweizer Gemeinden* [Data set]. opendata.swiss.
- [4] MeteoSwiss. (2026). *SwissMetNet open data, station Luzern (LUZ)* [Data set]. data.geo.admin.ch. Source: MeteoSwiss.
- [5] Fraunhofer ISE. (2026). *Energy-Charts: Public net electricity generation in Switzerland* [API]. api.energy-charts.info.

Methods and context

- [6] Shumway, R. H., & Stoffer, D. S. (2017). *Time series analysis and its applications: With R examples* (4th ed.). Springer.
- [7] Seabold, S., & Perktold, J. (2010). statsmodels: Econometric and statistical modeling with Python. *Proceedings of the 9th Python in Science Conference*.
- [8] Das, A., Kong, W., Sen, R., & Zhou, Y. (2024). A decoder-only foundation model for time-series forecasting. *ICML 2024*. arXiv:2310.10688.
- [9] Holmgren, W. F., Hansen, C. W., & Mikofski, M. A. (2018). pvlib python: A python package for modeling solar energy systems. *Journal of Open Source Software*, 3(29), 884.
- [10] Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and practice* (3rd ed.). OTexts.
- [11] United Nations. (n.d.). *Goal 7, target 7.2*. sdgs.un.org/goals/goal7
- [12] Swissgrid. (n.d.). *PV forecasting*. [swissgrid.ch](https://www.swissgrid.ch)

All links accessed on 8 Oct 2026. Figures marked “own count” are reproducible from [1].